complex model

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Hierarchical Block-Structured Bayesian Model

Let

$$y_i = \log(\text{SalePrice}_i)$$

denote the response for house i.

Likelihood

$$y_i \sim \mathcal{N}(\mu_i, \sigma)$$

Linear Predictor

$$\mu_i = \beta_0 + \alpha_{j[i]} + \eta_i^{(L)} + \eta_i^{(I)} + \eta_i^{(B)} + \eta_i^{(S)} + \eta_i^{(C)} + \beta_1 \cdot \text{GrLivArea}_i + \beta_2 \cdot \text{MasVnrArea}_i + \beta_3 \cdot \text{YearSinceRemodel}_i$$

Neighborhood Effect

$$\alpha_j \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha)$$

Lot Block Effect

$$\eta_i^{(L)} = w_1 \cdot \text{LotFrontage}_i + w_2 \cdot \text{LotShape}_i + w_3 \cdot \text{LandContour}_i + w_4 \cdot \text{LotConfig}_i$$

$$w_1, w_2, w_3, w_4 \sim \mathcal{N}(0, \tau_L)$$

$$\tau_L \sim \text{Half-Normal}(0, 1)$$

Interior Block Effect

$$\eta_i^{(I)} = w_5 \cdot \text{OverallQual}_i + w_6 \cdot \text{HeatingQC}_i + w_7 \cdot \text{FireplaceQu}_i + w_8 \cdot \text{GarageCars}_i + w_9 \cdot \text{CentralAir}_i + w_{10} \cdot \text{Electrical}_i$$

$$w_5, \ldots, w_{10} \sim \mathcal{N}(0, \tau_I)$$

$$\tau_I \sim \text{Half-Normal}(0, 1)$$

Basement Block Effect

$$\eta_i^{(B)} = w_{11} \cdot \text{BsmtQual}_i + w_{12} \cdot \text{BsmtExposure}_i + w_{13} \cdot \text{BsmtFinType1}_i + w_{14} \cdot \text{BsmtFinSF1}_i + w_{15} \cdot \text{TotalBsmtSF}_i$$

$$w_{11}, \ldots, w_{15} \sim \mathcal{N}(0, \tau_B)$$

$$\tau_B \sim \text{Half-Normal}(0, 1)$$

Structure Block Effect

$$\eta_i^{(S)} = \delta_{a[i]}^{(\text{MSSubClass})} + \delta_{b[i]}^{(\text{MSZoning})} + \delta_{c[i]}^{(\text{HouseStyle})} + \delta_{d[i]}^{(\text{Exterior1st})} + \delta_{e[i]}^{(\text{MasVnrType})} + \delta_{f[i]}^{(\text{Foundation})} + \delta_{g[i]}^{(\text{Condition1})}$$

$$\delta_a^{(\text{MSSubClass})}, \delta_b^{(\text{MSZoning})}, \delta_c^{(\text{HouseStyle})}, \delta_d^{(\text{Exterior1st})}, \delta_e^{(\text{MasVnrType})}, \delta_f^{(\text{Foundation})}, \delta_g^{(\text{Condition1})} \sim \mathcal{N}(0, \tau_S)$$

$$\tau_S \sim \text{Half-Normal}(0, 1)$$

Sale Condition Effect

$$\eta_i^{(C)} = \delta_{h[i]}^{(\text{SaleCondition})}$$

$$\delta_h^{(\text{SaleCondition})} \sim \mathcal{N}(0, \tau_C)$$

$$\tau_C \sim \text{Half-Normal}(0, 1)$$

Prior Distributions

Global Intercept

$$\beta_0 \sim \mathcal{N}(12, 1)$$

Regression Coefficients

$$\beta_1, \beta_2, \beta_3 \sim \mathcal{N}(0, 1)$$

Neighborhood Hyperpriors

$$\mu_\alpha \sim \mathcal{N}(0, 1)$$

$$\sigma_\alpha \sim \text{Half-Normal}(0, 1)$$

Noise Term

$$\sigma \sim \text{Half-Normal}(0, 1)$$